Early dynamic changes in pulse oximetry signals in preterm newborns with histologic chorioamnionitis

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Objective: **No reliable clinical markers of histologic chorioamnionitis (HCA), a major and often subclinical cause of prematurity leading to high neonatal morbidity and mortality, are available to date. Increasing evidence indicates myocardial dysfunctions in affected fetuses and newborns. We sought to assess the value of nonlinear dynamics from pulse oximetry signals in identifying affected newborns.**

Design: **Prospective case-control study.**

Setting: **Tertiary level neonatal intensive care unit, Brindisi Hospital.**

Patients and Intervention: **Pulse oximetry–derived signals (pulse rate, oxygen saturation, and perfusion index), recorded within the first 1.5 hrs of life, were analyzed for 110 very low-birth-weight infants, of whom 54 had histopathological evidence of HCA.**

Measurements and Main Results: **Four different time series parameters were determined for nonlinear dynamical (NLD) analysis. Significantly decreased Lempel-Ziv, Lyapunov largest exponent, and correlation dimension, with significantly increased Hurst values for heart rate and perfusion index ($p < .00001$), were observed in newborns with HCA. Heart rate Lempel-Ziv $\leq$0.218 showed 100% sensitivity (95% confidence interval, 98.8–100) and 100% specificity (95% confidence interval, 98.6–100) in distinguishing cases from controls, with positive and negative predictive values of 100% and 95.7%, respectively.**

Conclusions: **Our findings indicate that early autonomic tone balance abnormalities are present in newborns with HCA and suggest that early dynamic analysis of pulse oximetry signals could be useful in identifying affected infants. (Pediatr Crit Care Med 2006; 7:138–142)**

KEY WORDS: pulse oximetry; chorioamnionitis diagnosis; placenta diseases; Apgar score

Preterm birth occurs in about 5% to 7% of live births (12% in the United States) and is one of the most important risk factors for neonatal morbidity and mortality. Most of the serious associated problems occur in the 1% to 2% of the infants who are born before 32 completed wks' gestation and the 0.4% who are born before 28 wks' gestation (1). Chorioamnionitis (CA) is found in about 50% of very low-birth-weight (VLBW) infants and up to 80% of the extremely low-birth-weight (ELBW) infants. Because CA is a typical pathology of the second half of gestation and often a subclinical cause of preterm birth, it is related to increased neonatal morbidity and mortality, white matter brain damage, and cerebral palsy (2, 3). It effects are likely mediated by a fetal inflammatory response syndrome with multiple organ failure and dysfunction (4). CA can be considered a "natural" model for neonatal illness severity in high-risk newborns admitted to the neonatal intensive care unit (NICU) (5–11).

Although no early, noninvasive, validated, accurate markers of HCA in preterm newborns are available, recent research indicates that abnormal heart rate variability is associated with HCA in the term fetus (12, 13). Administration of endotoxin to an experimental newborn model has been shown to lead to an altered balance of autonomic tone, with reduced beat-to-beat heart rate variability (14). Emerging evidence indicates that nonlinear dynamic (NLD) analysis of vital rhythms may provide valuable practical diagnostic and prognostic information during the neonatal period (15–18). The pulsatile signal indexed against the nonpulsatile signal and expressed in percent (AC/DC $\times$ 100) is commonly referred to as the perfusion index (PI). In pulse oximetry, a constant amount of light is absorbed by skin, other tissues, and nonpulsatile blood, whereas a variable amount of light is absorbed by pulsating arterial inflow. PI is a scaled numeric value derived from the magnitude of the pulsations displayed on the plethysmographic waveform and is calculated as the percentage of pulsatile signal to nonpulsatile signal. Recently, PI has been shown to be a valid assessment tool for illness severity in both high-risk newborns and adults and to be associated with HCA (19–21), a major predictor of morbidity and mortality in VLBW newborns (11). Here, we tested the hypothesis that a shift in the fractal stochastic dynamics of pulse oximetry–derived pulse rate (PR) and PI toward periodicity (Fig. 1) is associated with HCA in a VLBW infant population.

METHODS

Subjects. We performed a case-control, prospective study of 110 VLBW infants (49 males and 61 females; gestational age, 28.6 ±

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2.2 wks; birth weight, 1005 ± 255 g), consecutively enrolled at the tertiary referral center for neonatal intensive care of the Brindisi Hospital (Brindisi, Italy) from January 1, 2002, to June 30, 2004. We obtained approval from our local human investigation committee and informed consent from parents of the infants before inclusion in the study. Placental examination was performed for all cases, and CA was defined according to previously described criteria (22). Pulse oximetry-derived heart rate (HR), oxygen saturation, and PI were monitored during the first day of life with a pulse oximeter (Masimo Set Radical, Masimo, Irvine, CA), with the sensor placed randomly on either of the feet, after verifying that the pulse wave was artifact-free, according to a previously described methodology (20). Further investigation is required to assess possible effects of instrument measurement variability on dynamical analysis. Illness severity in the first 1–12 hrs from birth was evaluated by several different neonatal severity scoring systems (Clinical Risk Index for Babies [CRIB], CRIB-II, Score for Neonatal Acute Physiology [SNAP] II, and SNAP Perinatal Extension [SNAPPE II]), as previously described (5–11). Pulse oximetry was performed by experienced nurses and neonatologists. Illness severity scoring was independently performed by two neonatologists, according to published protocols, while NLD analysis was carried out by mathematicians with experience in chaos theory application to biomedicine. In order to avoid possible bias in the placental histology interpretation, the presence or absence of HCA was assessed by experienced pathologists blinded to the NLD analysis results. Interrater coefficient of variation for NLD analysis was ≤0.5%, whereas interobserver variability was ≤1% for all the scoring systems used.

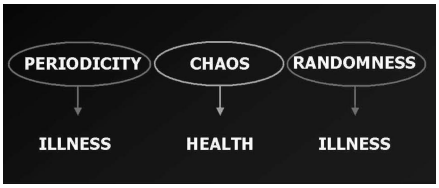

Figure 1. Fractal stochastic dynamics of pulse oximetry–derived signals: A hypothesis.

Nonlinear Dynamical Analysis. NLD analysis for HR and PI was carried out (by authors blinded to clinical data) in the few days following birth (median; range) and involved 1000 to 10,000 data point time series, recorded at 4-sec intervals, and the use of a commercially available software package (Chaos Data Analyzer [CDA] Pro, The Academic Software Library, NC State University, Raleigh, NC). The NLD parameters evaluated (relative Lempel-Ziv complexity [LZ], Lyapunov largest exponent [λ_i], correlation dimension [CD], and Hurst exponent [H]) are shown in Table 1.

The relative LZ is a measure of the algorithmic complexity of the time series. In the calculation, the CDA software converts each data point to a single binary digit according to whether its value is greater than or less than the median value (23). The largest Lyapunov (λ_i) exponent is an estimate of the rate at which trajectories in phase space diverge, reaching zero or assuming negative values for periodic data, weakly positive for chaotic data, and highly positive for uncorrelated random data (theoretically = infinite) (24). The correlation dimension (CD) was assessed by estimating the fractional dimension obtained by counting the data points inside hyperspheres of various radii centered on each data point in a reconstructed phase space, with use of a variation of the algorithm by Grassberger and Procaccia (25). We selected the minimal embedding dimension (ED) for which CD > 0, starting the analysis with ED > 4 and ending with ED < 11. Finally, the Hurst exponent (H) is a measure of the extent to which the data can be represented by a fractional Brownian motion (26). The value for H is <0.5 for anticorrelated series, >0.5 for positively correlated series, and 0.5 for random white noise series (i.e., Brownian noise). For example, using the CDA software, a sine function typically showed LZ = 0.04, λ_i = −0.001 ± 0.026, CD = 0.894 ± 0.624 (ED = 5), and H = 0.72, whereas randomly generated data showed LZ = 1.04, λ_i = 0.895 ± 0.031, CD = 4.4 ± 0.048, and H = 0.003 (n = 2000 data points, according to the algorithm by Press & Teukolsky) (27).

Statistical Analysis. Fit to normal distribution for these variables was assessed with the Kolmogorov-Smirnov test. Results were expressed as mean (SD) or median, range, and lower and upper 95% confidence intervals (CIs). Differences between groups were assessed with the Student's *t*-test or Wilcoxon's test, as appropriate. A two-sided $p < .05$ was judged statistically significant, and Bonferroni-corrected significance levels were used for multiple Student's *t*-tests. Sensitivity and specificity were evaluated with a receiver-operator characteristic (ROC) curve.

Table 1. Nonlinear dynamic parameters analyzed in pulse oximetry—derived time series

NLD Parameter	Description	Value/Range
LZ	A measure of the algorithmic complexity of the time series; the higher the LZ, the higher the complexity (i.e., randomness) of the series	0–1
λ_i	An estimate of the rate at which trajectories in phase space diverge	≤0 → periodic data • weakly positive → chaotic data • highly positive (theoretically = infinite) → uncorrelated random data
CD	Conceptually equivalent to fractal dimension → the higher the CD, the greater the complexity of the process	>0
H	A measure of the extent to which the data can be represented by a fractional Brownian motion	0–1 • if H = 0.5 (Brownian motion) → absence of long memory • if H < 0.5 → anticorrelated long memory process, • If H > 0.5 → correlated long memory process

NLD, nonlinear dynamic; LZ, relative Lempel-Ziv complexity; λ_i, Lyapunov largest exponent; CD, correlation dimension; and H, Hurst exponent.

RESULTS

On the basis of the placental histology results, 54 newborns (49.1%; 25 males and 29 females) were categorized as HCA-positive, and positive clinical criteria for CA were present in 19 cases (35.2%). HCA-positive infants showed significantly higher CRIB, CRIB-II, SNAP-II, and SNAPPE II scores ($p < .0001$; Table 2) than the HCA-negative population. Typical pulse-oximetry recordings are shown in Figures 2 and 3. The HCA-positive infants exhibited significantly decreased PI-LZ, PI-CD, PI-H, HR-LZ, HR-λ_i, and HR-CD values ($p < .0001$) and significantly increased PI-H and HR-H values ($p < .0001$), in comparison with HCA-negative control neonates (Table 3). In contrast, the trend for PI-λ_i did not reach statistical significance ($p = .059$). With use of the differences and SDs between the two groups, a power analysis of the observed differences indicated type I (α) and type II (β) levels of statistical error of 0.01 for all the examined variables, with the exception of PI-CD (type I and type II errors of 0.05). Sensitivity and specificity

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Table 2. Illness severity scores for very low-birth-weight infants, as a function of histologic chorioamnionitis

Neonatal Illness Severity Scoring System	Histologic Chorioamnionitis	Controls	*p* Value
CRIB	7 (6, 10)	3 (1, 4)	<.0001
CRIB-II	13 (12–14)	9 (8–10)	<.0001
SNAP-II	39 (29–45)	13 (5–17)	<.0001
SNAP-PE-II	68 (55–73)	68 (55–73)	<.0001

CRIB, Clinical Risk Index for Babies; SNAP, Score for Neonatal Acute Physiology; PE, Perinatal Extension.

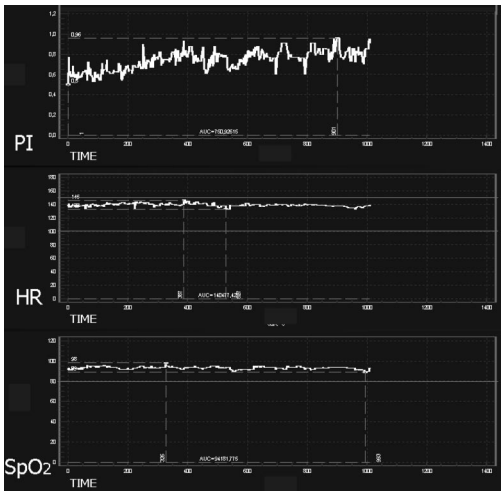

Figure 2. Typical pulse-oximetry (*SpO2*, %) recordings (about 1000 data points with 4-sec intervals, corresponding to about 67 mins) in case 1, a very low-birth-weight infant with histologic chorioamnionitis and remarkable reduction in heart rate (*HR*, beats per minute) variability (x-axis, time; y-axis units, perfusion index [*PI*, %]).

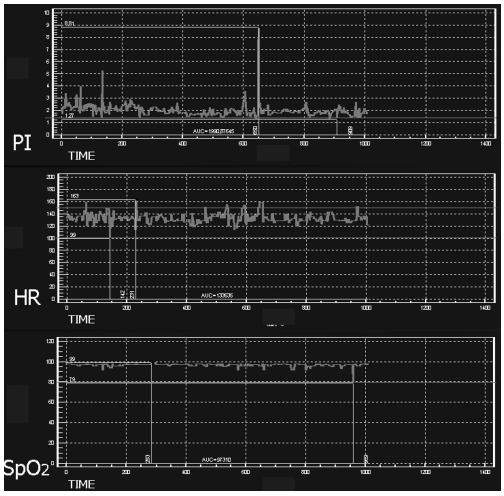

Figure 3. Typical pulse-oximetry (*SpO2*, %) recordings (about 1000 data points with 4-sec intervals, corresponding to about 67 mins) in case 2, a very low-birth-weight infant without histologic chorioamnionitis and with normal heart rate (*HR*, beats per minute) variability (x-axis, time; y-axis units, perfusion index [*PI*, %]).

of HR-LZ as a diagnostic test were 100% (95% CI, 98.8–100) and 100% (95% CI, 98.6–100) for cases and controls, with a cut-off value of ≤0.218. The positive and negative predictive values were 100% and 95.7%, respectively. The area under the ROC curve was 0.982. Differences in potential confounding factors between groups were nonsignificant ($p \geq .45$) and included gender, gestational age, birth weight, body temperature, and percentage of infants receiving oxygen supplementation.

DISCUSSION

These findings indicate that chaotic analysis of pulse oximetry signals can be applied to the early identification of HCA in VLBW newborns. Our prior research has generated several early markers for HCA in specific, selected high-risk neonatal categories (22, 28, 29) and identified HCA as a major predictor of morbidity and mortality among VLBW newborns.

In the past decade, several neonatal illness severity scores (NISSs) have been developed, including the CRIB (5), CRIB-II (6), SNAP (9), SNAP-PE (8), SNAP-II (10), and SNAPPE-II (10). However, controversy currently exists about the validity of the available NISS systems. In particular, potential limitations may arise from the concepts on which the currently available methods are based. First, treatment modalities have been shown to influence neonatal outcomes, as indicated by the report of increased mortality odds for inborn infants of <32 wks' gestation admitted to NICUs at night vs. during daytime (30) and the reduced incidence of a major adverse outcome for ELBW infants in a minimal handling environment (31).

Second, the potential effects of perinatal conditions on postnatal disease severity should be taken into account when evaluating the NISS. In particular, HCA represents an emerging challenge in perinatology. HCA, mediated by a fetal inflammatory response syndrome, is often subclinical and is associated with high rates of preterm birth. As a consequence, it leads to high fetal/neonatal morbidity and mortality (3, 4) and could be considered as a natural model for neonatal illness severity in high-risk newborns. To this regard, we have recently demonstrated that, after adjustment for potential confounders in a multivariable logistic regression analysis, HCA is an independent predictor of high illness severity in VLBW infants (11).

Finally, and most important, it should be noted that the calculation of risk assigned to a measured score is an epidemiologic tool and should not be used as a single patient prevision tool (32). These current limitations have stimulated a search for new, real-time indices of neonatal severity based on different concepts.

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Table 3. Comparisons of nonlinear dynamics variables from pulse oximetry signals in very low-birth-weight infants with and without histologic evidence of chorioamnionitis

Variable	Histologic Chorioamnionitis (n = 54)	Controls (n = 56)
Perfusion index		
Lempel-Ziv	0.36 ± 0.08	0.46 ± 0.07[a]
Lyapunov largest exp (λ_i)	0.39 ± 0.19	0.45 ± 0.09
Correlation-dimension	1.93 ± 1.01	2.81 ± 0.80[a]
Hurst	0.33 ± 0.08	0.26 ± 0.01[a]
Heart rate		
Lempel-Ziv	0.18 ± 0.04	0.46 ± 0.20[a]
Lyapunov largest exp (λ_i)	−0.25 ± 0.28	0.42 ± 0.13[a]
Correlation-dimension	1.31 ± 1.18	3.25 ± 1.15[a]
Hurst	0.36 ± 0.08	0.20 ± 0.02[a]

[a]$p < .00001$ vs. controls.
Values are mean ± SD.

We have previously assessed the potential value of skin colorimetry (33, 34), skin spectrophotometry (35), and pulse oximetry–derived PI (19, 21) in identifying different categories of higher-severity newborns.

The approach of the present study introduces the concept of real-time illness-severity prediction by exploiting the dynamic characteristics of pulse oximetry time series.

Several physiologic time series obtained in the cardiopulmonary system are inhomogeneous and nonstationary, fluctuating in an irregular and complex fashion. These fluctuations are often labeled as "noise" in the medical literature to distinguish them from the "true" signal of interest. The tool used in the present manuscript (NLD analysis of PI, PR, and pulse-oximetry variability) is to be considered totally new in that it is based on pulse oximeter–derived signals (pulse-to-pulse, oxygen saturation, and PI variability analysis) as opposed to electrocardiography-derived signals (R-R interval variability analysis), described in previous reports concerning illness severity and chaotic analysis.

Increasing evidence indicates that physiologic time series may carry important information about the structure and function of the heart and lungs. Examples include fluctuations in blood pressure, heart rate, respiratory rate, lung volume, ventilation and perfusion, tidal volume, end-expiratory oxygen and carbon dioxide levels, leukocyte counts, and blood flow (36–45). In living organisms, vital natural rhythms should be sufficiently regular to maintain constant function and sufficient flexibility in stress adaptation. The dynamic analysis of heart rate conveys relevant information regarding the health status of the newborn. In fact, heart-rate-characteristic measurements may be of clinical utility in estimating the risk of neonatal mortality in the NICU. Abnormal heart-rate characteristics of reduced variability and transient decelerations precede the clinical signs of sepsis and sepsis-like illness in neonates (15–18). We have recently observed that persistent fetal heart rate hypovariability as well as prolonged fetal bradycardia may be unrecognized signs of HCA (12, 13). Antepartum fetal monitoring with classic cardiotocography has been used for checking fetal status. Even by using the automatic cardiotocography analysis methods currently employed, pathologic fetal conditions can be detected. Recently, it has been observed that a multiparametric fetal heart rate analysis that includes spectral parameters from autoregressive models and nonlinear algorithms (approximate entropy) can identify pathologic fetal tracings (45). Cumulating evidence indicates that HCA is associated with a fetal inflammatory systemic response syndrome (4, 11, 46). Our research on acute thymic involution in newborns and fetuses with proven HCA is in line with this notion (22, 47–49). Although these data are suggestive of a similarity between the systemic inflammatory response syndrome of the fetus/newborn and that seen in children and adults, to date no information exists on this intriguing issue. An abnormal acute-phase response, along with increased proinflammatory cytokine secretion and decreased anti-inflammatory cytokine concentrations, is a shared feature. The HCA-associated fetal systemic inflammatory response (4) has been shown to lead to severe fetal cardiovascular compromise (50–52), and cardiac dysfunction results in dysregulation of cerebral blood flow as well as subsequent hypoxic-ischemic brain injury (53). HCA is a well-known risk for neonatal sepsis but is not obviously equivalent. For instance, this difference in terms of thymus histology concerning the HCA-associated acute thymic involution has been explored (21).

Our findings indicate the occurrence of an early loss in chaotic behavior for pulse-to-pulse variability in newborns with HCA, suggesting that placental inflammatory disease has an unrecognized impact on the neonatal autonomic regulation. HR is under direct neuroautonomic control and can be readily measured by noninvasive means. Previously, an abnormal chaotic behavior in interbeat variability has been shown to be an important predictor of heart failure in adult subjects (29). We speculate that the observed early abnormalities in HR and PI dynamics may be related to subclinical cardiovascular dysfunction in affected infants. Whether the observed abnormal autonomic tone is directly or indirectly related to the pathogenesis of the HCA-associated brain injury merits further study. Whether the observed changes in HR and PI dynamics are specific to HCA or, alternatively, are simply a reflection of higher illness severity in infants born with placental inflammatory disease needs to be clarified. Finally, we have shown that pulse oximetry signals can represent a valid source for NLD analysis. In light of our findings, we anticipate evaluating the validity of the NLD approach for identifying HCA during intrauterine life, monitoring its evolution, and possibly applying potential preventive and/or therapeutic strategies.

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